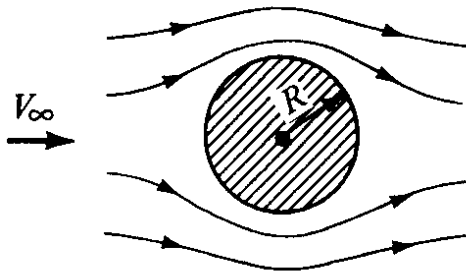


LIFTING FLOW OVER A CYLINDER

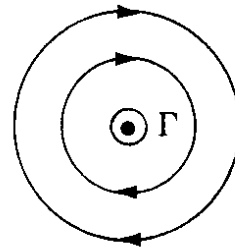
cylinder *spinning* about its axis

spinning



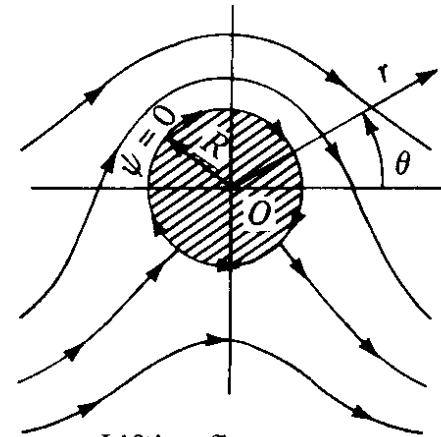
Nonlifting flow
over a cylinder

+



Vortex of
strength Γ

=



Lifting flow over
a cylinder

$$\psi_1 = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) \quad \psi_2 = \frac{\Gamma}{2\pi} \ln r + \text{const} \quad \text{Const} = -\frac{\Gamma}{2\pi} \ln R$$

$$\psi_2 = \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

$$\psi = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) + \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

LIFTING FLOW OVER A CYLINDER

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta$$

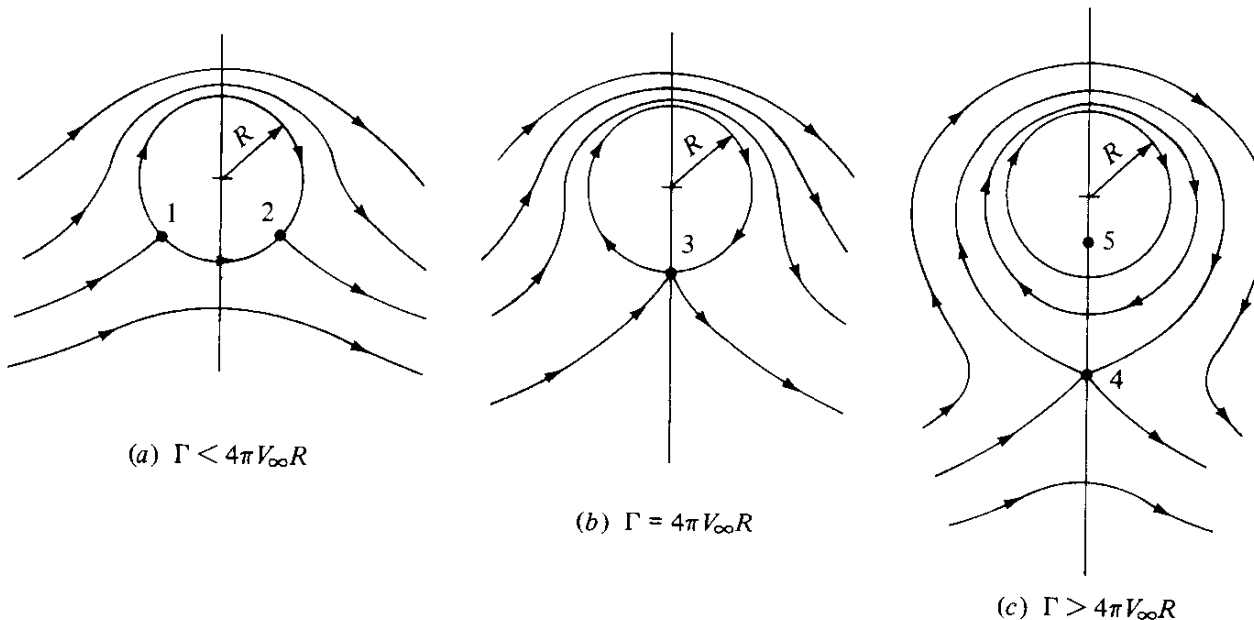
$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta - \frac{\Gamma}{2\pi r}$$

stagnation points in the flow, set $V_r = V_\theta = 0$



$$r = R$$

$$\theta = \arcsin\left(-\frac{\Gamma}{4\pi V_\infty R}\right)$$



Read relevant
discussions

Figure 3.33 Stagnation points for the lifting flow over a circular cylinder.

LIFTING FLOW OVER A CYLINDER

The velocity on the surface of the cylinder $V = V_\theta = -2V_\infty \sin \theta - \frac{\Gamma}{2\pi R}$

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 = 1 - \left(-2 \sin \theta - \frac{\Gamma}{2\pi R V_\infty} \right)^2$$

$$C_p = 1 - \left[4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \right]$$

$$c_d = c_a = \frac{1}{c} \int_{\text{LE}}^{\text{TE}} (C_{p,u} - C_{p,l}) dy \quad y = R \sin \theta \quad dy = R \cos \theta d\theta$$

$$c_d = \frac{1}{c} \int_{\text{LE}}^{\text{TE}} C_{p,u} dy - \frac{1}{c} \int_{\text{LE}}^{\text{TE}} C_{p,l} dy \quad c = 2R \quad c_d = \frac{1}{2} \int_{\pi}^0 C_{p,u} \cos \theta d\theta - \frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \cos \theta d\theta$$

$$c_d = -\frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta \quad \text{Substituting Equation (3.126) into (3.130)}$$

$$\int_0^{2\pi} \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin \theta \cos \theta d\theta = 0$$

$$c_d = 0$$

The lift on the cylinder

$$c_l = c_n = \frac{1}{c} \int_0^c C_{p,l} dx - \frac{1}{c} \int_0^c C_{p,u} dx \quad x = R \cos \theta \quad dx = -R \sin \theta d\theta$$

$$c_l = -\frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \sin \theta d\theta + \frac{1}{2} \int_{\pi}^0 C_{p,u} \sin \theta d\theta$$

$C_{p,l}$ and $C_{p,u}$ are both given by the same analytic expression.

$$c_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta$$

$$\int_0^{2\pi} \sin \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^3 \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta d\theta = \pi$$



$$c_l = \frac{\Gamma}{RV_{\infty}}$$

$$L' = \frac{1}{2} \rho_{\infty} V_{\infty}^2 2R \frac{\Gamma}{RV_{\infty}}$$

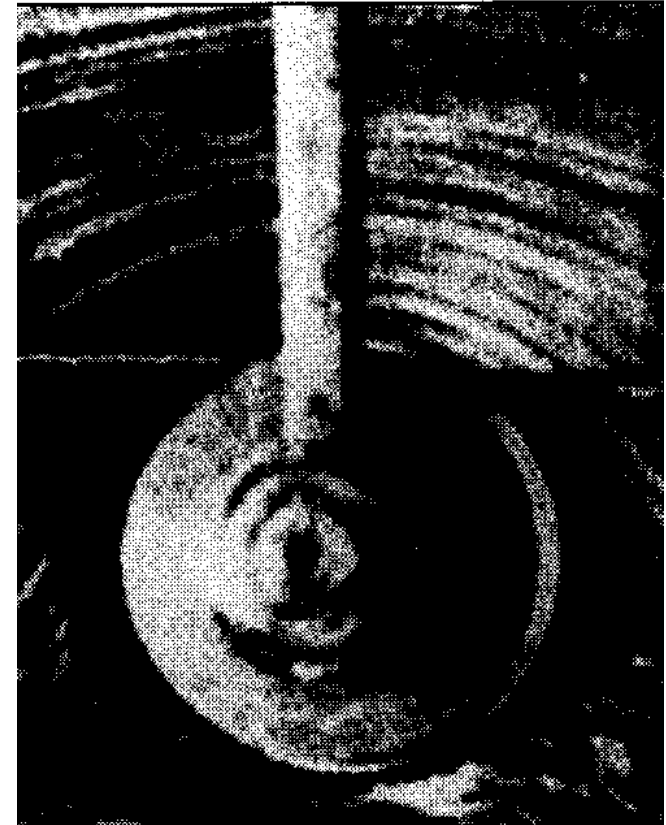
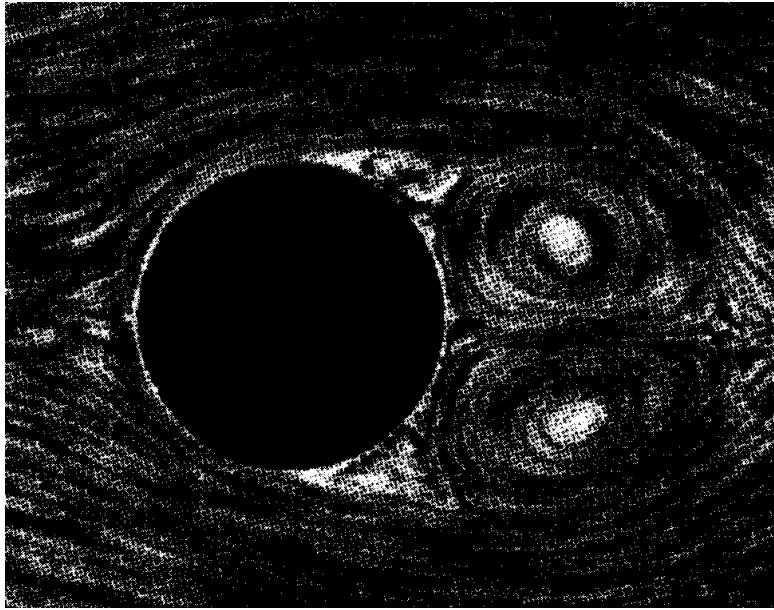
$$\boxed{L' = \rho_{\infty} V_{\infty} \Gamma}$$

Kutta-Joukowski theorem

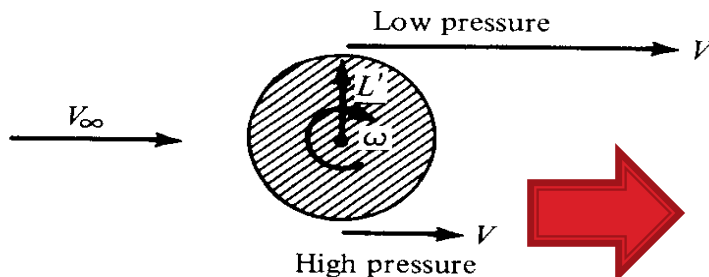
lift per unit span is directly proportional to circulation

The lift on the cylinder

Magnus effect,



These flow-field pictures were obtained in water, where aluminum filings were scattered on the surface to show the direction of the streamlines.
(a) Shown above is the case for the nonspinning cylinder. (Source: Prandtl)



(continued) These flow-field pictures were obtained in water, where aluminum filings were scattered on the surface to show the direction of the streamlines. (b) Spinning cylinder: peripheral velocity of the surface = $3V_{\infty}$. (c) Spinning cylinder: peripheral velocity of the surface = $6V_{\infty}$.

Figure 3.35

Creation of lift on a spinning cylinder.

Pressure difference between top/bottom

THE KUTTA-JOUKOWSKI THEOREM AND THE GENERATION OF LIFT

$$\Gamma \equiv \oint_A \mathbf{V} \cdot d\mathbf{s}$$

L' on the airfoil

$$L' = \rho_{\infty} V_{\infty} \Gamma$$

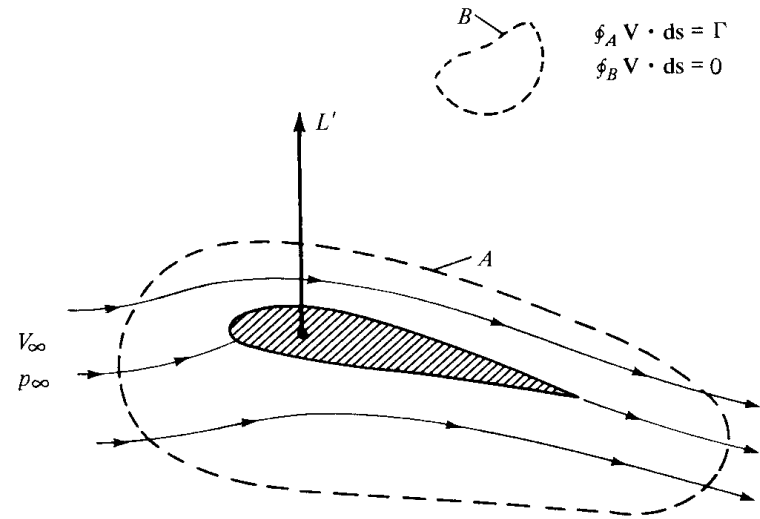


Figure 3.37 Circulation around a lifting airfoil.

The general derivation of Equation (3.140) for bodies of arbitrary cross section can be carried out using the method of complex variables.

curve *not* enclosing the origin, we obtain from Equation (2.137) the result that $\Gamma = 0$.

The Kutta-Joukowski theorem is simply an alternative way of expressing the *consequences* of the surface pressure distribution